

Architectural Blueprint for an AI-Driven Embodied Dance Pedagogy System

SECTION 1: Core Teaching Loop Design

The fundamental efficacy of an artificial dance instructor relies on its ability to emulate the cognitive and sensorimotor feedback loops utilized in human-led dance pedagogy. Dance is a complex embodied activity requiring the continuous integration of proprioceptive, visual, and auditory information¹. In a digital environment, the system must parse incoming kinematic data, identify critical deviations, and deliver precise, non-overlapping verbal cues that respect the limitations of human cognitive load.

1a. The Real-Time Correction Engine

To simulate a live instructor, the real-time correction engine must dictate what to say, when to say it, and how to deliver it within a 500–1000 millisecond window during an active attempt.

a) Priority Queue System During the execution of a complex movement sequence, multiple joints may deviate from the reference trajectory simultaneously. Emitting corrections for all errors concurrently induces cognitive overload and violates the principles of spaced practice². The system employs a hierarchical priority queue to determine the single most critical joint to correct.

The queue operates on a weighted sorting algorithm. First, joints falling within the active lesson phase (e.g., "arms only") receive a 2.0x weight multiplier to enforce the current focus area. Second, the system applies a proximal-to-distal priority matrix. Motor control initiates at the core and radiates outward; correcting a proximal joint (like the shoulder) often organically resolves distal deviations (like the wrist)². Therefore, shoulders and hips receive a higher base priority than elbows and knees. Finally, a historical weighting mechanism is applied. If a joint was corrected in the previous computational cycle, it receives a temporary priority boost to facilitate closed-loop feedback, ensuring the user completes the correction before the system abruptly shifts its focus to a new body part. Beginner modes prioritize gross motor alignments (hips, shoulders), while advanced modes unlock distal tracking (wrists, ankles) for fine-tuning.

b) Timing & Delivery Rapid ballistic movements cannot be corrected mid-flight due to the human cognitive processing delay. Therefore, real-time cues must be targeted at continuous actions or strategically queued. Delivery relies on the window.speechSynthesis API, which must be tightly controlled to prevent audio overlap⁴. The system utilizes the speaking and pending attributes to monitor the audio thread. If an error is detected but the API is currently speaking, the new correction is dropped to prevent queue buildup, unless its severity exceeds a 40° deviation threshold. In the case of severe errors, speechSynthesis.cancel() is invoked to interrupt the current phrase and immediately inject the urgent cue⁴. To accommodate the rapid pace of movement, phrases delivered mid-routine are truncated to ultra-short imperatives (e.g., "Left arm up!") to fit within a 1-second delivery window, while longer analytical phrases are reserved for post-attempt reviews. Voice dynamics are modulated programmatically: severe

errors trigger a higher pitch (1.2) and rate (1.2) to convey urgency, while mild corrections utilize a calm, normalized tone (pitch 1.0, rate 1.0)⁶.

c) Positive Reinforcement Motor memory consolidation relies heavily on positive reinforcement when a target kinematic state is achieved⁸. The system tracks improvement by maintaining a Simple Moving Average (SMA) of the Mean Absolute Error (MAE) for each joint over a rolling 10-frame window. If a previously flagged joint exhibits a >20% reduction in its MAE compared to its historical baseline over three consecutive windows, a VERIFYING_FIX state triggers an immediate praise utterance. To ensure every attempt ends on a positive note, the session concludes with a "sandwich feedback" algorithm: the system first highlights the joint with the lowest overall error, gently notes the primary area for improvement based on the highest average deviation, and ends with a generalized encouraging statement.

d) The Correction Phrase Database

To prevent robotic repetition, the phrase database utilizes a nested data structure mapping 8 core joints across 3 severity thresholds. The system selects phrases using a bag-of-words shuffle algorithm drawn without replacement until all variations are exhausted.

Joint	Severity	Mid-Movement (1-2s)	After-Action (3-4s)	Praise When Fixed
Left Shoulder	Mild (15-25°)	"Watch the left shoulder." / "Left shoulder placement." / "Check left shoulder."	"Your left shoulder drifted slightly out of alignment. Keep it engaged."	"Good shoulder placement." / "Left shoulder looks great."
	Moderate (25-40°)	"Left shoulder up!" / "Lift the left shoulder." / "Raise left side."	"You need to elevate your left shoulder to match the horizontal line."	"Much better on the left shoulder!" / "That is the right height."
	Severe (>40°)	"Raise that left shoulder!" / "Left shoulder dropping!" / "Lift the left!"	"Your left shoulder completely dropped. Focus on keeping it elevated."	"There it is! Perfect shoulder height." / "Excellent correction."

Right Shoulder	Mild (15-25°)	"Level the right shoulder." / "Right shoulder check." / "Watch the right."	"Pay attention to the slight dip in your right shoulder."	"Nice right shoulder control." / "Right side is solid."
	Moderate (25-40°)	"Right shoulder higher!" / "Push the right shoulder up." / "Lift right!"	"Your right shoulder needs more elevation on that extension."	"Great adjustment on the right shoulder." / "Right shoulder fixed."
	Severe (>40°)	"Lift your right shoulder!" / "Right shoulder is lost!" / "Pull up right!"	"You are losing the frame on your right side. Lift from the joint."	"Yes! Right shoulder is locked in." / "Beautifully leveled."
Left Elbow	Mild (15-25°)	"Soften left elbow." / "Left elbow line." / "Check left arm."	"Your left elbow is a bit too stiff, try to soften the line."	"Good elbow shape." / "Nice line on the left."
	Moderate (25-40°)	"Bend left elbow!" / "Shape the left arm." / "Curve the left!"	"You are locking your left elbow. Keep a natural, soft bend."	"That is the right elbow angle!" / "Perfect bend."
	Severe (>40°)	"Unlock left elbow!" / "Bend it left!" / "Left arm too straight!"	"Your left arm is completely straight. It needs a distinct bend."	"Excellent! Left elbow is fixed." / "Much better arm shape."
Right Elbow	Mild (15-25°)	"Right elbow check." / "Right	"Keep an eye on the angle of	"Nice shape on the right arm." /

		arm shape." / "Soften the right."	your right elbow."	"Right elbow is good."
	Moderate (25-40°)	"Straighten right elbow!" / "Extend right arm." / "Reach right!"	"You need more extension through your right elbow on that beat."	"Much better extension there." / "Great reach."
	Severe (>40°)	"Reach right fully!" / "Straighten the right!" / "Right arm out!"	"Your right arm is too bent. Reach fully outward to complete the line."	"Beautiful reach on the right side!" / "Full extension, perfect."
Left Hip	Mild (15-25°)	"Square left hip." / "Left hip placement." / "Watch left hip."	"Your left hip is rotating slightly outward. Keep it squared."	"Hips are squared nicely." / "Good hip alignment."
	Moderate (25-40°)	"Turn left hip in!" / "Left hip forward." / "Square up left!"	"Bring your left hip forward to square your pelvis to the front."	"Good hip rotation." / "Left hip is back in place."
	Severe (>40°)	"Left hip is open!" / "Square your hips!" / "Close the left hip!"	"Your left hip is completely open, breaking the line. Square up."	"Perfectly squared hips now!" / "Excellent pelvic control."
Right Hip	Mild (15-25°)	"Right hip check." / "Watch right hip." / "Level	"Notice the slight drop in your right hip. Keep it level."	"Right hip looks solid." / "Good level on the hips."

		the right hip."		
	Moderate (25-40°)	"Lift right hip!" / "Right side up." / "Pull up right hip!"	"You are sinking into your right hip. Pull up from the core."	"Much better weight distribution." / "Great hip lift."
	Severe (>40°)	"Pull right hip up!" / "Don't sink right!" / "Lift the right side!"	"You are collapsing onto your right hip. Engage your core to lift it."	"Great lift on the right side!" / "Hips are perfectly level."
Left Knee	Mild (15-25°)	"Soften left knee." / "Left knee check." / "Watch the left leg."	"Your left knee is slightly stiff on the landing. Soften it."	"Good bend in the left knee." / "Nice soft landing."
	Moderate (25-40°)	"Bend left knee!" / "Plie left." / "Sink into the left!"	"You need a deeper bend in your left knee for stability."	"Nice depth on the left side." / "Great plie."
	Severe (>40°)	"Left knee deep!" / "Bend the left!" / "Don't lock the left!"	"Your left leg is totally stiff. You must bend the knee to absorb the movement."	"Excellent plie on the left!" / "Much safer knee bend."
Right Knee	Mild (15-25°)	"Right knee line." / "Check right leg." / "Right knee direction."	"Make sure your right knee tracks over your toes."	"Right knee alignment is good." / "Right leg looks strong."
	Moderate (25-40°)	"Straighten right leg!" /	"You need to fully extend	"Great extension on

		"Extend right knee." / "Push through right!"	your right knee on this beat."	the right." / "Perfect straight leg."
	Severe (>40°)	"Lock right knee!" / "Straighten the right!" / "Straight right leg!"	"Your right leg is bent when it should be a perfectly straight line."	"Beautiful straight right leg!" / "Excellent extension."

e) The Correction State Machine

A finite state machine (FSM) governs the transition of pedagogical attention per joint. The system defaults to the MONITORING state, continuously evaluating the FastDTW-aligned angle differences. When a joint crosses the >20° threshold, it transitions to DETECTED_ERROR. The phrase selection algorithm fires, synthesizing speech, and the state shifts to CORRECTING. A strict cooldown timer (2,000ms) initiates to prevent nagging. Once the cooldown expires, the system enters VERIFYING_FIX. If the joint's SMA improves by >20%, the state shifts to PRAISE and then back to MONITORING. If the user fails to fix the error after 3 consecutive loops through the CORRECTING state, the system transitions to FRUSTRATION_AVOIDANCE, suppressing further cues for this specific joint during the current attempt to preserve the user's morale, logging the persistent error for the After-Action Review instead.

SECTION 2: The Lesson Plan Architecture

The pedagogical framework breaks continuous movements into digestible, physically isolated tasks, mirroring the cognitive load theory applied in physical skill acquisition¹.

2a. Progressive Lesson Flow

Phase 0 — TEACH THE MOVE Before practice begins, the system extracts the 3 to 5 most structurally significant keyframes from the video chunk. Algorithmically, this is achieved by calculating the temporal local maxima of the overall kinetic energy. By computing the sum of the Euclidean displacements of all body landmarks between frames,

$$E_k = \sum \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$
, the system isolates moments of peak extension or directional shifts⁹. These extracted keyframes are rendered as static Canvas 2D overlays, accompanied by descriptive synthesized voiceovers (e.g., "First, the arm sweeps up").

Phases 1 to 5 — Isolation and Combination During Phase 2 (Arms Only) and Phase 3 (Legs Only), the visual isolation of non-active body parts is achieved by manipulating the globalAlpha property of the Canvas 2D context during the skeleton rendering loop¹¹. Inactive bones are rendered at 0.15 opacity, naturally drawing the user's foveal vision to the focus area without entirely removing the body's grounding context. The scoring algorithms adjust dynamically; in

Phase 2, the denominator for the total score is restricted to joints 11-16 (shoulders, elbows, wrists)¹². If a user consistently scores above 85% in Phase 4 (Combine at 0.75x), a dynamic routing system allows them to bypass isolation phases in subsequent chunks, proceeding directly to full speed.

2b. The Counting System

Precise auditory synchronization is non-negotiable. Using standard JavaScript `setInterval` for a metronome results in severe timing jitter due to the single-threaded event loop¹³.

The counting system relies on the Web Audio API's hardware clock (`AudioContext.currentTime`). Following the "A Tale of Two Clocks" scheduling paradigm, a lightweight JavaScript interval checks ahead and schedules audio buffer nodes in the future directly on the audio hardware thread, guaranteeing sample-accurate playback¹³. When the `playbackRate` shifts to 0.5x, the chunk duration doubles. The scheduling algorithm divides the original reference chunk duration by 8 to establish the native beat interval, then multiplies this interval by $(1/\text{playbackRate})$. Counts 1 and 5 utilize distinct audio samples with higher dynamic compression and pitch to simulate a teacher's natural accentuation. Visually, a central pulsating ring operates via CSS transforms, triggered by a `requestAnimationFrame` loop that polls `AudioContext.currentTime` to ensure the visual expansion synchronizes precisely with the audio transient¹⁶.

2c. The After-Action Review (AAR) Screen

Following an attempt, raw frame-by-frame data is condensed into actionable visual intelligence. Rendering overlaid skeletons efficiently on Canvas 2D requires spatial normalization. The

system calculates a translation vector $T = \text{Root}_{ref} - \text{Root}_{user}$ (using the mid-hip coordinate) and applies `ctx.translate()` to ensure both skeletons share a unified origin¹⁸.

For joints exceeding the 25° error threshold, indicator arrows are rendered. The rotation angle of the arrow is derived from the spatial vector between the reference joint and the user's joint using `Math.atan2(dy, dx)`, rotating the Canvas context to draw a high-contrast red arrow pointing toward the correct spatial position²⁰. Score progression is displayed via a lightweight SVG line chart that visualizes the delta between attempts. Improvement is defined strictly by the percentage reduction in MAE compared to the previous attempt. If the user improved their arms but degraded their legs, the system explicitly acknowledges this divergence: "Your arm path improved by 15%, but we lost some bend in the knees. Let's balance them both."

SECTION 3: Technical Architecture

3a. The Correction Pipeline

To maintain the strict 15fps performance budget required for real-time webcam analysis, the architecture isolates the computer vision processing from the main UI thread using `Web Workers`²².

The main thread captures the webcam via `requestAnimationFrame` and draws the

HTMLVideoElement to an OffscreenCanvas. This canvas context generates an ImageBitmap via transferToImageBitmap(), which is passed to the Web Worker via postMessage. This zero-copy transfer circumvents the computationally expensive serialization of image arrays²⁴. Inside the worker, MediaPipe Pose operates via a WebAssembly/WebGL delegate, extracting 33 normalized 3D landmarks¹². The worker then calculates the 8 critical joint angles using inverse tangent mathematics:

$$\theta = |\text{atan2}(y_3 - y_2, x_3 - x_2) - \text{atan2}(y_1 - y_2, x_1 - x_2)| \quad ^3.$$

Because users naturally drift off-beat, comparing the user's instantaneous frame to the reference video's current frame is flawed. The worker applies a FastDTW (Dynamic Time Warping) algorithm to align the user's recent sequence buffer with the reference timeline, calculating the true spatiotemporal error²⁶. These spatial errors are transmitted back to the main thread, where the Correction Engine enforces a debouncing throttle before invoking SpeechSynthesis.speak().

3b. The Phrase Selection Algorithm

The algorithm filters and weights joint scores to determine the most pedagogically sound correction.

```
TypeScript
function selectCorrection(
  currentScores: JointScore[],
  stateMachine: Map<JointId, JointState>,
  focusArea: 'arms' | 'legs' | 'full',
  speechSynthesis: SpeechSynthesis
): void {
  // Prevent overlap unless critical severity triggers an interrupt
  if (speechSynthesis.speaking) return

  // 1. Filter and weight current scores based on pedagogy
  let weightedScores = currentScores.map(score => {
    let weight = 1.0
    if (isProximal(score.jointId)) weight *= 1.5 // Prioritize core/shoulders/hips
    if (isInFocusArea(score.jointId, focusArea)) weight *= 2.0 // Enforce lesson phase
    if (stateMachine.get(score.jointId).recentlyCorrected) weight *= 1.2 // Follow-through
    return { score, weightedError: score.error * weight }
  })

  // 2. Check for praise conditions first
```



```

const improvedJoint = checkImprovement(stateMachine, currentScores);
if (improvedJoint) {
    triggerPraise(improvedJoint);
    return;
}

// 3. Find worst joint
weightedScores.sort((a, b) => b.weightedError - a.weightedError);
const worst = weightedScores[0];

// 4. Execute correction if above baseline threshold
if (worst.error > 20) {
    const phrase = getUniquePhrase(worst.jointId, worst.severity);
    speak(phrase);
    updateStateMachine(worst.jointId, 'CORRECTING');
}

```

If the user drops out of the frame entirely, the MediaPipe visibility score falls below a confidence threshold (0.5), triggering a specialized error handler: "Step back, I can't see your full body"²⁵.

3c. Performance Budget

Standard DTW computation possesses an $O(N^2)$ time complexity, which would rapidly consume the CPU budget. The system mitigates this by utilizing the FastDTW approximation, reducing complexity to $O(N)$ by projecting the alignment path from lower resolutions and employing a Sakoe-Chiba constraint band²⁸. Re-scoring and DTW alignment occur every 3rd frame (~200ms interval). This cadence provides sufficient granularity for fluid visual feedback while halving the computational overhead. The memory footprint for storing a 5-minute session's historical score array is approximately 1.8 MB, comfortably managed by browser garbage collection.

SECTION 4: User Experience & Psychology

4a. Teacher Personality Design

A dance teacher's effectiveness is profoundly linked to their pedagogical personality¹. The system features an adaptable personality matrix based on the user's skill level.

- **Encouraging/Beginner:** High kinematic tolerance thresholds (30° deviation before cueing). Utilizes expressive, metaphorical language. ("Feel the stretch through your fingertips," "Let your shoulders float down").
- **Strict/Advanced:** Low kinematic tolerance thresholds (15°). Highly specific, anatomical

cues focusing on technique. ("Lock the right elbow," "Square the pelvic floor"). The system automatically shifts from the beginner to the advanced profile as the user's historical MAE improves across sessions.

4b. Motivation & Progress

To build a sense of momentum, the system implements localized operant conditioning. If the user maintains a "green" score ($<15^\circ$ error) across all active joints for 4 consecutive beats, a subtle, highly compressed audio chime plays—a non-verbal reinforcement that does not interrupt the music. Session progress is reinforced during transition screens: "You've mastered 3 out of 8 chunks. Let's tackle the bridge." Milestone moments trigger specialized speech variations: "Your arm score is the highest it has been all week!"

4c. Error Handling UX

Systemic failures must never masquerade as user failures. If the webcam connection is lost, the skeleton immediately fades to gray, video playback pauses, and the teacher states, "I've lost your camera feed. Don't worry, your progress is saved. Step back into the frame when you're ready." If a user gives up mid-attempt and stops moving, the system detects a near-zero velocity profile, pauses the routine, and asks, "No problem. Do you want to slow it down, or try a different move?"

SECTION 5: What Else Can We Add?

5a. Pre-Practice Warm-Up

A guided 2-minute mobility sequence primes the neuromuscular system and calibrates the camera's tracking volume³². The system tracks simple joint isolations (e.g., neck rolls, shoulder shrugs, hip circles) using MediaPipe, verifying the user's anatomical range of motion prior to the complex routine.

5b. Difficulty Progression Across Chunks

Chunks are algorithmically sorted by a Kinematic Complexity Score. This score is calculated by integrating the average joint velocity and the degree of asymmetrical limb movement over the chunk's duration. The curriculum automatically orders the routine from the least to most complex segments, dynamically reordering subsequent chunks based on the user's real-time success rate.

5c. Cross-Session Memory

Using the browser's IndexedDB API, the system serializes a session summary object containing the `worst_performing_joints` array and specific spatial weaknesses. Upon initiating the next session, the teacher loads this profile, greeting the user with context: "Welcome back. Last time we struggled with the left arm extensions. Let's see if we can perfect them today."

5d. Dance Style Adaptation

Different genres demand distinct pedagogical priorities³³. For classical forms like Ballet, scoring weights shift heavily to the hips, knees, and ankles (points 23-28) to enforce turnout and posture¹². Conversely, popular styles in India, such as Bollywood, place immense emphasis on expressive mudras (hand gestures), rapid body isolations, and thumkas (hip thrusts)³⁵. In the Bollywood configuration, the scoring algorithm exponentially increases the weight on the wrist and distal extremity landmarks (MediaPipe points 15-22). Correction phrases adapt to the regional vernacular: "Sharpen that hand gesture," "More bounce in the hip," or "Keep the mudra clear"³⁵.

5e. Musicality Coaching

Matching a spatial pose is secondary if the user is entirely off-beat¹. To measure musicality, the system extracts the 1D velocity profile of the user's movement and performs a cross-correlation with the reference dancer's velocity profile. The lag index of the cross-correlation peak dictates temporal accuracy. If the lag is consistently negative, the teacher provides timing-specific corrections: "You're rushing the beat. Wait for the 3."

5f. Social / Mirror Mode

In physical studios, students rely on mirrors to align themselves with the instructor³⁸. The app implements a true mirror mode by applying CSS transform: scaleX(-1) to both the webcam feed and the reference video. This eliminates the cognitive load of spatial translation, allowing the user to map their physical right arm directly to the teacher's visible "right" arm on the screen.

5g. Teaching Different Skill Levels

Beginner levels suppress the scoring overlay entirely, replacing it with a simplified heat map, while the teacher instructs, "Just try to follow the rhythm, don't worry about perfect accuracy." Advanced levels display real-time angular degrees and velocity vectors, demanding precision on highly subtle expressions and musicality.

SECTION 6: If YOU Were Building The Dance Teacher From Scratch

If unconstrained by legacy code, the ideal pedagogical engine is built feature-by-feature as a fully immersive, state-driven application.

6a & 6b. Greenfield Design Specifications

1. The Lesson Structure Engine

What it is: Orchestrates the macro-flow from warm-up to cool-down dynamically based on user biometrics and success rates. **Why it matters:** Human teachers do not force a rigid curriculum if a student is struggling; they pivot. **How to build:** Implemented as a hierarchical state machine using XState. Transitions between states (WARMUP, TEACH, DRILL, REVIEW) are

guarded by algorithmic condition checks (e.g., `averageScore > 60%`). **Edge cases:** If the session duration exceeds 20 minutes and kinematic scores begin dropping rapidly (indicating physical fatigue), the engine overrides the curriculum and forces a `COOL_DOWN` state.

Pedagogy: Avoids overtraining and injury². **Priority/Effort:** P0 / 5 days.

2. The Real-Time Voice Corrector

What it is: Synthesizes low-latency, context-aware verbal feedback during active movement.

Why it matters: Provides closed-loop motor feedback without requiring the user to break visual focus on the reference¹. **How to build:** A priority queue fed by the FastDTW output.

Calculates `worst.error = max(abs(user.angle - ref.angle))`. Uses SpeechSynthesis to deliver a phrase from the database. **Edge cases:** If the Speech API hangs (a known Chrome issue), a watchdog timer monitors `speechSynthesis.pending` and forcefully restarts the API if execution stalls for `>2000ms`⁴. **Pedagogy:** Corrects proprioceptive drift instantly. **Priority/Effort:** P0 / 7 days.

3. The Counting System

What it is: A continuous audio/visual metronome providing the "5, 6, 7, 8" backbone. **Why it**

matters: Establishes the temporal framework essential for dance³⁸. **How to build:**

AudioContext lookahead scheduling loop scheduling pre-recorded vocal samples exactly on the beat interval. **Edge cases:** Backgrounding the browser tab pauses `requestAnimationFrame`. The AudioContext hardware clock continues natively, instantly resyncing the visual UI ring upon tab focus¹³. **Pedagogy:** Enforces rhythm over pure spatial imitation. **Priority/Effort:** P0 / 3 days.

4. The Progress Tracker

What it is: Visualizes session-over-session and attempt-over-attempt kinematic improvement.

Why it matters: Visual proof of progression is the strongest driver of user retention.

How to build: A React charting component plotting the normalized FastDTW distance over time, using SVG paths with cubic bezier smoothing to hide minor frame-to-frame noise.

Edge cases: If a user performs catastrophically worse on one attempt, the smoothing algorithm truncates the downward spike to avoid demoralization.

Pedagogy: Fosters a growth mindset.

Priority/Effort: P1 / 2 days.

5. The After-Action Review (AAR)

What it is: A post-mortem visual and verbal breakdown occurring immediately after a chunk attempt. **Why it matters:** Consolidates learning while muscle memory is fresh². **How to build:**

A React modal triggered on the `CHUNK_COMPLETE` state. Fetches the cached `worst_performing_joints` array, overlays skeletons via `ctx.translate`, and renders directional `Math.atan2` vectors pointing out deviations¹⁸. **Edge cases:** Extreme deviation where the user is facing the wrong way; the Euclidean distance check suppresses the overlay to prevent visual spaghetti. **Pedagogy:** Bridges the gap between feeling the movement and seeing the result.

Priority/Effort: P0 / 4 days.

6. The Warm-Up Generator

What it is: Guides the user through a 2-minute mobility sequence prior to the routine. **Why it matters:** Primes the neuromuscular system, reduces injury risk, and covertly verifies the camera's tracking volume³². **How to build:** A hardcoded 3-step sequence (neck, shoulders, hips) verified via simple boundary checks on MediaPipe joint coordinates. **Edge cases:** User skipping warm-up. System warns them but allows bypass, reducing the kinematic intensity of the first chunk. **Pedagogy:** Standard physical education practice. **Priority/Effort:** P2 / 2 days.

7. The Difficulty Scaler

What it is: Adapts video playback rate and tolerance thresholds dynamically.

Why it matters: Keeps the user in the optimal "flow state" of learning.

How to build: PID controller logic tracking the MAE. If the user scores <40% twice sequentially, the controller output drops playbackRate to 0.5x and increases the error threshold to 30°.

Edge cases: User insists on full speed despite failing; the scaler offers a manual override button.

Pedagogy: Scaffolding instructional support.

Priority/Effort: P1 / 3 days.

8. The Correction Memory

What it is: Recalls historical weaknesses across different days to personalize instruction.

Why it matters: Makes the AI feel like a dedicated, long-term coach.

How to build: Store a JSON blob in IndexedDB keyed by userId. Upon load, inject historical worst joints into the priority queue with a 1.2x weight multiplier.

Edge cases: User deletes local storage. The system gracefully degrades to a blank slate without crashing.

Pedagogy: Emphasizes long-term skill retention.

Priority/Effort: P1 / 2 days.

9. The Focus Director

What it is: Forces the user to perfect one mechanism by visually dimming the rest of the body.

Why it matters: Reduces cognitive load for complex multi-limb coordination¹. **How to build:**

Manipulate CSS masking and Canvas globalAlpha during the requestAnimationFrame loop. Set

error thresholds for non-focus joints to ∞ so they are ignored by the scoring engine¹¹. **Edge**

cases: User moves non-focus limbs wildly; the system ignores it entirely, strictly adhering to the focus area. **Pedagogy:** Task simplification strategy. **Priority/Effort:** P0 / 1 day.

10. The Motivation Engine

What it is: Injects gamification, milestones, and praise based on micro-improvements.

Why it matters: Sustains engagement through the frustrating phases of motor learning.

How to build: Event listeners hooked to the scoring pipeline. Emits confetti and triggers specialized SpeechSynthesis buffers when the consecutive_green_frames counter exceeds a

threshold.

Edge cases: Rapidly oscillating scores triggering spam praise. Fixed via a 5000ms debounce on positive reinforcement.

Pedagogy: Operant conditioning.

Priority/Effort: P2 / 2 days.

11. The Musicality Coach

What it is: Evaluates temporal precision (rhythm) independently of spatial accuracy (pose).

Why it matters: Hitting a pose off-beat is incorrect in dance³⁸. **How to build:** Extract the 1D velocity profile of the user. Perform cross-correlation with the reference velocity. The lag index dictates rushing vs dragging²⁶. **Edge cases:** User stands perfectly still. Cross-correlation fails; handled by checking variance before calculation. **Pedagogy:** Trains the ear, not just the body.

Priority/Effort: P1 / 4 days.

12. The Style Adaptor

What it is: Modifies scoring tolerances and vocabulary based on the specific dance genre.

Why it matters: Hip-hop requires bounce; Bollywood requires precise mudras; Ballet requires rigid posture³⁵. **How to build:** A configuration object injected at runtime. BollywoodConfig multiplies the scoring weight on distal extremity landmarks (MediaPipe 15-22) by 3.0 and loads a genre-specific phrase database. **Edge cases:** Misidentified style; allows manual override in the settings menu. **Pedagogy:** Respects the cultural and technical nuances of diverse art forms. **Priority/Effort:** P2 / 3 days.

13. The Pose Breakdown

What it is: Extracts and explains critical moments of a move before practice begins. **Why it matters:** Provides a cognitive map of the sequence⁸. **How to build:** Calculate local maxima of joint velocity (first derivative of displacement). Extract those exact video frames to a hidden Canvas, export as Data URIs, and display in the Phase 0 UI⁹. **Edge cases:** Very smooth, slow movements without clear velocity peaks. Handled by falling back to sampling at 25%, 50%, and 75% of the chunk duration. **Pedagogy:** Establishes declarative knowledge before procedural execution. **Priority/Effort:** P1 / 3 days.

14. The Mirror Mode

What it is: A skeleton overlay that reverses orientation, acting as a digital mirror. **Why it**

matters: Eliminates the cognitive translation of "their right is my left"³⁸. **How to build:** CSS transform: scaleX(-1) applied to the video container. **Edge cases:** Asymmetrical choreography designed specifically for stage right/left. The app provides a toggle to disable mirror mode.

Pedagogy: Industry standard spatial learning technique. **Priority/Effort:** P0 / 1 day.

15. The Mistake Highlighter

What it is: Visual arrows and glowing lines demonstrating the delta between the user and the reference. **Why it matters:** Pinpoints exactly how to fix the spatial error. **How to build:**

Calculate the vector difference between the user's joint and reference joint. Use `ctx.setLineDash()` to draw a glowing path from the current position to the target position¹⁸.

Edge cases: Occlusion. If a limb is hidden behind the body, suppress the highlight to prevent erratic visual artifacts. **Pedagogy:** Visual augmented feedback. **Priority/Effort:** P0 / 2 days.

6c. What Am I Missing? (5 Advanced Features)

1. Tactical Silence (Cognitive Load Accumulator)

- *What it does:* Intentionally mutes all feedback for a period, letting the user dance uninterrupted.
- *Why it matters:* Constant feedback creates dependency. Humans need moments to process and fail safely without judgment¹.
- *How to build:* Implement a `cognitive_load` integer. Each verbal cue adds a value. If the accumulator exceeds a threshold, silence all non-critical feedback for 15 seconds, allowing the variable to decay.

2. Proprioceptive Questioning

- *What it does:* Instead of telling the user what they did, it asks them what they felt.
- *Why it matters:* Forces the student to internalize the physical sensation of correct technique².
- *How to build:* During the AAR, if the system detects a massive improvement, it pauses and displays a prompt: "You fixed your shoulder alignment. Could you feel the stretch in your back?"

3. Freeze-Frame Physical Adjustment

- *What it does:* The digital equivalent of a teacher physically moving a student's limb.
- *Why it matters:* Some errors require absolute static correction before resuming motion.
- *How to build:* The video pauses mid-movement upon detecting a severe error. A vector arrow appears on the user's arm. The app refuses to resume playback until the user physically moves their arm into the designated "green zone" on the screen.

4. Breathing Synchronization Cues

- *What it does:* Inserts programmed audio cues ("Inhale," "Exhale") mapped to movement expansion and contraction.
- *Why it matters:* Breath control is fundamental to stamina and movement fluidity in dance¹.
- *How to build:* Metadata timestamps added to the chunk data structure triggering specific, breath-focused `SpeechSynthesis` utterances.

5. Asymmetrical Feedback Adaptation

- *What it does:* Identifies if a user struggles more with their left or right side and adapts the lesson flow.
- *Why it matters:* Everyone has a dominant side. A real teacher drills the weak side heavier.
- *How to build:* Accumulate MAE separately for left-side joints vs right-side joints. If the variance exceeds 30%, dynamically insert an extra drill phase focusing solely on the weaker side.

6d. The Startup Sequence (First 2 Minutes)

To establish authority and trust immediately, the first two minutes are heavily orchestrated.

- **0:00 - 0:15 (Calibration):** The user grants camera permissions. The screen displays a glowing silhouette. The synthesized voice speaks calmly: "Hi, I'm your AI dance coach. Let's calibrate." The user is guided to step back until all 33 MediaPipe landmarks turn green, ensuring the tracking volume is secure²⁵.
- **0:15 - 0:30 (Trust Building):** The AI asks the user to raise their right arm. The system validates the movement, drawing a glowing line on the arm, and plays a success chime. The user immediately understands the AI can "see" and evaluate them accurately.
- **0:30 - 1:00 (Analysis):** The user selects a choreography video. The system displays a highly technical progress bar: "Analyzing kinematics... Extracting keyframes... Mapping rhythm." This transparent loading phase establishes the system's analytical depth.
- **1:00 - 1:30 (Deconstruction):** The screen splits. The AI speaks: "I've broken this routine into 4 learnable chunks. Let's look at the first move." Phase 0 initiates, showing the 3 extracted keyframes sequentially, demystifying the complex choreography.
- **1:30 - 2:00 (Execution):** "Let's try the arms slowly. Follow my count. 5, 6, 7, 8..." The music plays at 0.5x, the visual ring pulses perfectly on beat, the legs dim, and the user makes their first guided attempt, receiving their first real-time correction.

SECTION 7: Implementation Priority

1. Build Order

- *P0 (Core Infrastructure):* Web Worker MediaPipe pipeline, atan2 angle extraction, Canvas 2D skeleton rendering. Without this foundational tracking, the application cannot function²⁴.
- *P1 (Alignment & Scoring):* FastDTW implementation to align sequences temporally, followed by the threshold-based scoring logic²⁶.
- *P2 (The Teacher):* Web Audio API counting scheduler, SpeechSynthesis cue database, and the Priority Queue algorithm.
- *P3 (UX & Polish):* Lesson plan states, UI animations, AAR visual overlays, difficulty scaling.

2. Quick Wins

- **Audio Counting:** Implementing a flawless Web Audio API metronome requires minimal code but adds massive perceived value immediately, grounding the app in authentic dance culture¹⁶.
- **Keyframe Extraction:** Pulling out local maxima to show static poses requires minimal math (simple array derivative) but makes the pedagogical engine look highly intelligent³⁹.

3. Hard Problems

- **DTW Alignment at Scale:** Running sequence alignment every frame will lock the main thread. Optimizing the sliding window DTW to operate smoothly inside a Web Worker requires deep algorithmic tuning²⁸.
- **Skeleton Overlay Normalization:** Accurately mapping a 5'2" user's skeleton onto a 6'2" reference dancer's skeleton requires complex affine transformations and bone-length

normalization to avoid rendering visual spaghetti.

4. The Moat

- The true moat is the **Correction Phrase Selection Algorithm combined with Temporal Alignment**. Competitors can easily draw static skeletons on a screen using raw MediaPipe. An engine that knows exactly when a user is off-beat versus physically incorrect, manages cognitive load, and speaks an anatomically accurate, encouraging correction in real-time is profoundly difficult to replicate.

SECTION 8: Specific Technical Solutions

1. The correction timing problem: Speech overlap is prevented by wrapping SpeechSynthesis in a Mutex-style manager class. When speak() is called, it checks window.speechSynthesis.speaking or .pending. If true, the request is dropped. To align corrections with musical phrasing, the manager queues the utterance and fires it exactly 500ms *before* the next downbeat. It uses AudioContext.currentTime to sync the speech execution, ensuring the user hears the cue (e.g., "Left arm up") precisely as they initiate the next musical phrase⁴.

2. The focus-lock problem:

When focusing on arms, the leg skeleton must not disappear entirely; it drops to 0.2 opacity (ctx.globalAlpha = 0.2). A disappearing skeleton breaks the user's kinesthetic grounding. During this phase, the leg score is entirely omitted from the mathematical average denominator. The UI dynamically replaces the leg score with a "-" or "N/A" to clearly communicate the omission, ensuring the user isn't penalized for lower body positioning.

3. The speed transition problem: The audio metronome operates independently of the HTML5 video player, driven by a scheduleNote() loop checking AudioContext.currentTime against nextNoteTime¹³. When the playbackRate shifts from 0.5x to 0.75x, the secondsPerBeat variable is dynamically recalculated: $\text{secondsPerBeat} = (60.0 / \text{nativeBPM}) / \text{playbackRate}$. The user perceives the counting as naturally speeding up proportionally with the music, maintaining flawless synchronization.

4. The improvement detection problem: A strict frame-to-frame comparison is highly susceptible to MediaPipe jitter. The architecture employs a Simple Moving Average (SMA) over a 10-frame window (~330ms at 30fps). Improvement is strictly defined as: $\text{Current_SMA} < (\text{Previous_SMA} - 5^\circ)$ sustained for at least 3 consecutive windows. This smooths out algorithmic noise while introducing only a negligible ~300ms lag, well within the optimal threshold of human perception for positive reinforcement³⁸.

5. The phrase variation problem:

A static lookup table quickly induces repetitive fatigue. The system utilizes a nested token-replacement template: "{Action} {Target} {Direction}". Where {Action} = ["Push", "Lift", "Raise"], {Target} = ["your left arm", "that left side", "the arm"], and {Direction} = ["higher", "up"]. The manager uses a pseudo-random number generator with a "bag-of-words" shuffle algorithm (drawn without replacement) to guarantee that the same generated variation is never spoken twice in a single session.

6. The emotional tone problem: A "nagging" tone results from high pitch, fast rate, and repetitive phrasing. To convey a "coaching" warmth, the SpeechSynthesisUtterance parameters

are modulated programmatically: the pitch is set to 0.9 (slightly deeper, more grounded), and rate is set to 1.1 (energetic but not rushed)⁶. When a praise state is triggered, pitch temporarily bumps to 1.1 to simulate genuine excitement.

7. The skeleton overlay problem: Directly overlaying highly divergent poses creates confusing visual overlap. The system implements a Dynamic Spatial Threshold. The Euclidean distance between the normalized user root (hip) and reference root is calculated. If the delta

$D > 0.15$ (roughly 15% of the frame width), the direct overlay link is broken²⁶. The system stops rendering the user's limb over the reference limb. Instead, it transitions to a side-by-side view, drawing a translucent "ghost" trail connecting the user's current limb position to the target destination in their own frame, effectively turning an overlapping error into a clear visual path for correction.

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